

BEOPS

Belgrade Evidence Observatory

What exists, how it was made, what Belgrade turned out to have, what the literature leaves open, how it should look, how it should think, and what it could become — for a conference paper, a public presentation, and a commercial product.

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with Svemir — the authors' local AI infrastructure, on this work carried by Claude (Anthropic, model claude-fable-5-1) in a Cowork session, signing as a verified contributor at the authors' request. Contributor, not author: journal policy assigns authorship to people and responsibility to the editor of record; in git the same truth is the trailer Co-Authored-By: Claude Fable 5.1 on every commit this mind made.

Every number in this document was checked against files on disk or against a page that was actually opened on 8–9 September 2026; anything not checked is marked as such. Section 1 is in Serbian; the rest is in English because the paper and the public site will be English-first with a Serbian edition.

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1 • Sažetak

BEOPS je opservatorija Beograda zasnovana na dokazima: sistem koji sakuplja samo ono što grad javno i zakonito objavljuje, pamti za svaku vrednost *ko* ju je rekao, *kad* je izmerena (ako se to uopšte zna), *kad* je objavljena i *kad* je primljena — i koji odsustvo podatka beleži kao podatak, a nikad kao nulu. Rečenica projekta glasi: „**Ovo je ono što nam je Beograd rekao, kad nam je rekao, i gde je začutao.**”

Stanje 9. septembra 2026, provereno na fajlovima: registar od **189 izvora** (110 sondiranih, 3 sakupljena, 9 sa zabeleženim odbijanjem, 12 sa potvrđenim odsustvom), **167 izvora sa sačuvanim dokazom dozvole** (robots.txt, zaglavlja, licenca — sa hešom; 221 hvatanje), jedan zatvoren pilot (OBS-001, trka 10K, 10 od 13 termina, 270 vrednosti, izvor ne objavljuje vreme merenja), **kolektor koji od 8. septembra radi bez čoveka** svakih pet minuta (SEPA HVD API — 32 beogradske stanice, Sensor.Community — oko 145 građanskih senzora, Parking servis — 27 lokacija, šest novinskih feedova — samo naslov, link i vreme objave), 74 automatska testa, sistem reda pod kojim može da radi više umova, dve studije prikaza nad stvarnim redovima i registar organa (malih lokalnih modela) sa unosom za svaki po uzoru na algoritamske registre Amsterdama i Helsinkija.

Šta smo saznali o gradu: Beograd ima više instrumenata nego što objavljuje. Zvanična mreža za vazduh objavljuje čas merenja i, od maja 2026, API koji sam navodi svoj pravni osnov (EU uredba o skupovima velike vrednosti) — ali taj isti API označava lokalno vreme kao UTC, što smo izmerili prvog dana. Parking servis objavljuje brojeve bez ikakvog

vremena. Grad Beograd na nacionalnom portalu ima 7 skupova podataka, Novi Pazar 67. Kamere se ne smeju zakonito koristiti (sve vode do izdavača koji to zabranjuju); jedine zakonite „kamere“ su sateliti. Zvuka nema. Seizmička mreža je formalno otvorena, a arhiva prazna. GTFS u realnom vremenu postoji — samo za Google. Srbija nema izuzetak za text-and-data-mining ni u nacrtu zakona iz marta 2026, pa BEOPS stoji na dozvoli i licenci, ne na izuzetku — što je, pokazalo se, jača pozicija.

Šta literatura ostavlja otvoreno (provereno 8. 9. 2026): (1) kako zapisati i nacrtati vrednost čije je vreme merenja nepoznato — nijedan standard (OMS/ISO 19156:2023, SensorThings 2.0, OWL-Time) nema „nepoznato vreme“; (2) odsustvo kao prostorno agregiran zapis prve klase sa vizuelnom gramatikom; (3) prikaz starosti podatka u javnom prikazu — samo blogovi proizvođača; (4) verifikacioni sloj za izvore koje otkriju AI agenti; (5) empirija pokrivenosti čula u Srbiji. Ono što nije rupa i ne sme se preterivati: razlika prognoza/merenje, PROV-O, kritika digitalnih blizanaca, LLM agenti za pretragu skupova.

Šta predlažemo za dalje, ovim redom: sedam dana neprekidnog sakupljanja pre bilo kakve nove tvrdnje; kičma projekta van jednog diska (GitHub mirror je spreman kao skripta, dokazi ostaju lokalno); registar sa mašinskim poljima umesto proze; dnevni izveštaj o starosti i odsustvu kao prvi javni proizvod; javni sajt na beops.rs, dvojezičan, sa monologom grada kao glavnim interfejsom i mapom koja pulsira samo kad je instrument stvarno pročitano; organ uma od više malih lokalnih modela koji *izvode*, nikad ne mere, i čija je svaka rečenica označena; rad sa jednim doprinosom po radu (predlog: Places and Technologies 2027 i AGILE 2027 za konferencije, *Data & Policy* za časopis bez naknade). Kritike koje sigurno dolaze — „grad nije računar“, „otvoreni podaci nisu dovoljni“, „pa šta?“, „jedan čovek“, „AI je izmislio citate“, „skrejpovanje je nezakonito“ — imaju odgovore, i svaki je u poglavlju 11 sa onim što *ne smemo* da tvrdimo.

2 • Executive summary

BEOPS (Belgrade Evidence Observatory) is a small, deliberately slow system that collects only what Belgrade publishes lawfully and openly, and stores every value with its provenance — source, measurement time (when known), publication time, reception time, unit and permission — treating absence as a record and never as a zero. It is not a digital twin, not a smart-city platform and not a UN-Habitat observatory; it is an *evidence ledger with time semantics*, built by two academics on one university PC with a zero-dependency stack.

State (verified 9 Sept 2026). 189 source records; 167 with stored permission evidence (221 captures); one closed descriptive pilot (OBS-001); a collector running unattended since 8 Sept (12 configured sources, nine enabled, first day: SEPA 2 477 rows/h window filtered to 32 Belgrade stations, ~300 citizen-sensor rows per 10 min, 27 parking lots per 15 min, ~300 headlines per 30 min); 74 offline tests; two rendering studies over real rows; an organ registry with the first organ built and, by design, silent while the local model daemon is down. Two defects were found and fixed during the external review (a provenance index that called three permitted sources refusals; a pilot task that never deleted itself), and both are written up as corrections C-010 and C-011 — the project's method is that its own mistakes are data.

Findings about the city. Belgrade's official air network publishes measurement hours and, since May 2026, a keyless high-value-dataset API that cites its own legal basis — and labels local time as UTC, a defect measured on our first day. The parking operator publishes counts with no time at all. On the national portal the City of Belgrade has 7 datasets, Novi Pazar 67. No camera over Belgrade may lawfully be harvested; the only lawful cameras are satellites. No public sound exists. The seismic network is formally open and effectively empty. Real-time transit exists — for Google only. Serbia has no text-and-data-mining exception, and the March 2026 draft copyright law does not add one; BEOPS therefore stands on licence and captured permission, which is the stronger position.

Contribution. Three things are genuinely open in the 2024–2026 literature: a per-observation semantics and rendering for *unknown measurement time* (no standard has it); *absence* as a spatially aggregated first-class record with a visual grammar; and *data age* in public displays. A fourth — a verification layer for sources discovered by AI agents — is novel in application. Everything else BEOPS does is standard practice and must be cited as such.

Next. Seven unattended days before any new claim; the backbone off a single disk; machine-readable registry fields; a daily age-and-absence report as the first public product; a bilingual site at beops.rs whose main interface is the city's monologue beside a map that pulses only when an instrument was actually read; an organ of mind made of small local models that derive and never measure; one contribution per paper.

3 • What BEOPS is, and is not

BEOPS observes one city, Belgrade, at three grains only — the whole city, its larger units, and broad zones — and stops before the micro level: no individual, no building, no street as a unit of public analysis. It connects sources and hosted services that already exist; it builds no geocoder, no city index and no new infrastructure platform. It began on 5 September 2026 as a research-first foundation after an earlier application prototype was archived as material, not runtime.

The five hard rules (CONTRIBUTING.md; each has a test that fails when it is broken)

1. **No invented measurement.** A number that no source reported does not exist. Interpolation, imputation and back-filling are forbidden; a model may only *estimate*, and an estimate is drawn and stored as one.
2. **Received is not measured.** Every row keeps three times apart: the source's measurement time (`phenomenonTime`, or `null` with a reason), the source's publication time (`resultTime`), and our reception time (`receivedTime`). They never collapse into one.
3. **Forecast is not measurement.** Predictions and estimates are a different state with a different stroke, never a filled shape with a crisp edge.
4. **Missing is not zero.** A missing value is `null` with `resultQuality="missing"`; a silent slot is a silent slot; a dead sensor, an opted-out publisher and a never-collected source are three different recorded states.
5. **Every claim carries source, time, space, unit and permission.** Permission is bytes on disk (the source's own `robots.txt`, headers and licence page, hashed), captured *before* any collector is written, and an unknown is never a permission.

The five states

State	What it is	The lie if confused	Stroke · ground · colour
Observed	A source reported this value and its time	—	solid · clean · signal (chromatic)
Untimed	The value is exact; its measurement time is unknown	A stale number read as current	solid · age band fading left, no left edge · signal at reduced chroma
Estimated	A model or rule derived it	A derivation read as a measurement	dashed · clean · structure colour
Forecast	A statement about the future	A prediction read as a reading	dotted, quantile dots · clean · structure colour
Unavailable	Nothing came	Absence read as zero	a positive hollow mark on the baseline · hatched cell · state grey

The consequence is the design doctrine: **only observation may be chromatic**. The amount of colour on screen is the amount of real measurement in the picture; a bad day looks grey without anyone writing "data quality: poor".

What BEOPS is not

Not a digital twin (it refuses to synthesise a city it has not measured; see Borbach et al. 2025, Moroni 2025); not a smart-city platform (it deploys no sensors, sells no optimisation, holds no person-level data); not a UN-Habitat Local Urban Observatory (an indicator programme with a different governance); not a news archive (headlines are quoted short and the article is never kept); not real-time (its fastest source refreshes in minutes, and the word "real-time" is banned from its headlines because the parking source proved it cannot be honoured).

4 • What exists on 9 September 2026 — verified inventory

Component	State	Evidence
Repository D:\Svemir\!Projekti\Beops	28 commits since 5 Sept; no remote yet (a public export script exists, see §14); 1 782 tracked files; 74 offline tests green (npm test)	git log; test run 8 Sept 23:10 local
Order system	CONTRIBUTING.md (claim before write, status line, immutable evidence, what a number may claim), CLAIMS.md append-only, research/README.md as the index that decides which document wins, KNOWLEDGE.md in five sections, LOG.md, RECNİK.jsonl	files
Source registry	189 records: probe_ok 110 · primary_page 28 · lead 20 · no_coverage 10 · opted_out 9 · collected 3 · needs_decision 3 · dead 2 · token/restricted/account/blocked 1 each. Reuse rights verified for 12 of 189	research/SOURCE_REGISTRY.json
Provenance	167 sources with stored evidence, 221 captures; 153 access checks passed, 7 refused, 4 awaiting decision, 3 incomplete, 25 undocumented (after C-010)	research/08-provenance/INDEX.md, generated 8 Sept
Pilot OBS-001	Belgrade 10K race, 5 Sept: 10 of 13 hourly slots, 27 lots, 270 displayed values, no measurement time published; pre-registered zone group falsified and reported as such	research/observations/10k-2026-09-05/IZVESTAJ.md
Live collector	tools/collect_daemon.py under Windows task Beops_Collect every 5 min (S4U, runs unattended; verified firing, last result 0). 12 sources configured, 9 enabled. Transport curl.exe/Schannel (OS certificate store). Data in data/live/ outside git	scheduler query; runtime/collect-tick.log
First live day	SEPA HVD: 2 477 rows per 2-h window, filtered to 32 Belgrade stations, deduplicated; Sensor.Community: 290-310 rows per 10 min from ~145 sensors; Parking: 27 lots per 15 min, time unknown; news: 304 headlines in the first pass (Tanjung 164, Danas 50, Danubeogradu 50, Gradnja 30, Beta 10; Studio B connection reset)	receipts on disk

Organs	research/ORGANS.json (algorithm register): <i>news-sorter</i> built and tested (4 tests), silent by design while the model daemon is down; <i>series-watch</i> and <i>daily-voice</i> specified	tools/organ_news.py
Design studies	traka-live.html (time as primary object) and monolog-puls.html (bilingual monologue + pulsing map) over the live snapshot; earlier state-alphabet.html	research/05-design/studies/
Literature	93 + 12 + 32 + 24 + 7 + 7 + 7 + 8 verified entries across eight dossiers, plus 63 entries from the 8 Sept review	research/04-bibliography/, _trail/SPOLJNI_PREGLED_2026-09-08_ISTRAZIVANJE.md
Legal frame	Serbian primary texts quoted; no TDM exception relied on; official gazette outside copyright (Art. 6(2)); seven named opt-outs honoured; six questions marked for a lawyer	research/07-legal/
Paper	Serbian pre-paper V2 with structure, methods audit, claim ledger and venue audit; historical manuscript frozen; conference deadline (20 Aug) passed; nothing sent	research/06-paper/

Honest verdict on the inventory. The project is a catalogue and legal frame of an observatory that has, since 8 September, started to become an observatory. It has one day of continuous series. Every claim in this document about "what Belgrade does" rests on sources that were opened; every claim about "what BEOPS has" rests on files that were listed, and the two must not be confused.

5 • Method

5.1 Discovery — where the finds actually came from

Almost nothing of value was found by "searching the web for the thing". Seven mechanisms account for every non-trivial discovery, and they are now the discovery protocol (research/01-programme/HYPOTHESES_2026-09-06.md):

1. **Catalogues leak endpoints on domains you would never guess.** The best source in the registry, SEPA's HVD API, lives at `opendata.kosava.cloud`; it surfaced only as a resource URL inside a `data.gov.rs` dataset. Extracting every resource URL from the 7064 harvested resources gave 62 distinct hosts, 61 not the portal.
2. **Front-ends name their back-ends, and back-ends have standard doors** (ArcGIS `/rest/services`, udata `/api/1/`, CKAN `/api/3/`, GeoServer `GetCapabilities`, DCAT

/catalog.jsonld).

3. **Files exist where no index lists them** — the 86 road-count workbooks were recovered by treating one file URL as a template.
4. **The obligation creates the data — search the law, not the web.** Retail price lists exist because a regulation obliges them; the SEPA API exists because of Regulation (EU) 2023/138 on high-value datasets and says so as an ELI URI.
5. **Registers are duties, and duties are public by law.**
6. **Checking permission is itself discovery** — robots.txt files name AI agents and declare Article 4 reservations.
7. **Ask "instrument or world?"** before reporting a change: Serbia's road-count series fell because counters fell from 317 to 227, not because traffic did.

A domestic lesson sits above all seven: for a city outside the EU, **search national institutions before global platforms**. Two rounds through world open-data networks found that the famous ones stop at the EEA38 border; the instruments were at home — RHMZ's radar inside the city, SEPA's stations, the seismic station BEO running since 1918, ten named streets with traffic counters reporting to one named office.

5.2 Verification and the permission capture

tools/legal_capture.py stores, for one source at one moment: the site's robots.txt exactly as served (a 404 is stored too — under RFC 9309 the absence is itself a statement); the machine verdict of the robots parser run separately for our own user-agent and for `*`; the response headers of the exact URLs we will read, with `X-Robots-Tag`, `Content-Signal`, `Content-Usage`, `TDM-Reservation` and `TDM-Policy` pulled out by name; the licence or terms page in its original bytes; and a manifest with SHA-256 of each. The verdict is three-valued — permitted, refused, unknown — and unknown never renders as permitted: the tool's first run manufactured seven false permissions because a TLS failure yielded `None` and `None` is not `False` evaluated to `True` (C-001). The generated index lists every registered source without a capture under *not yet documented*, visibly, as the work queue.

The collector identifies as `Beops-Research-Collect/1.0`. There is no mode that disguises who is asking; a proposal for a collector not identified as an AI agent was refused and recorded as refused, because Pleiades' robots.txt names `ClaudeBot` and `anthropic-ai` with `Disallow: /` while `User-agent: *` permits everything — a collector under any other name reads as allowed, and choosing such a name is circumvention.

5.3 The legal frame, in one paragraph

Serbia has not transposed Directive (EU) 2019/790; there is no text-and-data-mining exception and no press publishers' right, and the draft Law on Copyright and Related Rights put to public consultation on 16 March – 6 April 2026 transposes the orphan-works and collective-management directives and the Marrakesh Treaty but, according to three independent reports (the draft itself was not opened), contains no TDM provision. BEOPS therefore relies on none: it processes, derives and discards, quotes short and attributed (Art. 49), keeps only what a licence or an open-data regime permits, and requests through the freedom-of-information law what matters. The Copyright Act's Art. 6(2) excludes laws, regulations and official materials of public bodies from copyright entirely, so the *Službeni*

list grada Beograda is the cleanest unused corpus in the country. Personal data: the scientific-expression exemption is ZZPL Art. 88; news feeds are kept as metadata at city level and no person-level record is retained or derived. Named opt-outs (Beobuild; Zoomer, Insajder, eKapija under Cloudflare Content-Signal; Politika, Vreme, Nedeljnik fencing their feeds; three camera aggregators) are honoured and recorded so that nobody rediscovers them next year and quietly starts collecting.

5.4 The role of AI minds, and the human ledger

Discovery, cataloguing, capture, literature search and drafting were done by AI agents (Claude and Codex sessions inside Svemir) under one human coordinator, in bounded lanes with claims (`CLAIMS.md`) and countersignature. Three rules make this defensible in front of a reviewer: every citation is verified by a human-directed check at the primary source and carries its reading level (full text, abstract, metadata, indexed excerpt); every source record carries the hash of its permission evidence, so a third party can re-verify each of the 189 records without re-running any agent; and the agent's failures are written up as corrections, not deleted. *Reproducibility is of the ledger, not of the discovery*: any third party can re-verify every record; nobody, including us, can reproduce the search that found them, and the paper must say so. The published evidence for why this matters is recent: 2026 audits report link validity above 94 % but factual accuracy of 39–77 % in LLM deep-research agents, and a twelve-fold rise in fabricated references in two years (Nature, 1 Apr 2026; Retraction Watch, 7 May 2026).

5.5 Corrections as method

`research/08-provenance/CORRECTIONS.md` holds eleven entries (Appendix D). Two are from the 8 September external review: C-010, the provenance index called three permitted sources refusals because it treated the mere presence of an `X-Robots-Tag` header as a refusal (`index, follow` on Kontur's CC BY dataset was listed under "said no"); C-011, the pilot's scheduled task never deleted itself because `cmd.exe` saturates twelve-digit numbers at 32 bits, so `GTR` was never true, and the task fired 69 empty times after the pilot closed. Both were fixed the same day, the fixes have tests, and the general rules ("a stored signal and a verdict on it are two different things"; "a schedule's end is verified at the scheduler, not in the script meant to end it") are now in the canon. The project's stance is that a provenance system that hides its own failures is worth nothing, because the one thing it must prove is that it does not quietly say untrue things.

6 • Belgrade's senses

6.1 The open layer — what version 1 stands on, asking nobody

The rule since 5 September is that v1 depends on nobody: no letters, no purchases, no waiting for permission — only what is lawfully open today, exhausted. On that rule the following is collected or collectable now:

Sense	Source (id)	Cadence	Measurement time?	State on 9 Sept
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Air, official	SEPA HVD API, <code>opendata.kosava.cloud</code> (S146): 87 stations nationally, 32 in the Belgrade bbox, 15 components incl. BTEX; hourly preliminary means; keyless; DCAT-AP HVD catalogue; legal basis as ELI URI	hourly	yes — interval per hourly mean; labelled local time as Z (defect, §7)	collecting since 8 Sept
Air, official, verified	SEPA verified daily archive on <code>data.gov.rs</code> (S06), XLSX 2011–2024, SODL	annual	yes (daily)	captured; the same measurement in two declared states, preliminary hourly and verified daily
Air, city institute	beoeko.com (S05), 31 stations, hourly, "unverified hourly values"	hourly	yes	probed; scrape-only; the only way to catch a broken official sensor, but must be compared station by station first
Air, citizen	Sensor.Community (S04): ~145 sensors within 25 km (PM, T, RH, P); Klimerko (map only); PurpleAir; WAQI mirror of SEPA	2.5–5 min	yes (UTC per record)	collecting since 8 Sept
Parking	Parking servis (S10): displayed free spaces, 27 lots, coordinates via operator's map links	unknown; page says nothing	no — no Last-Modified, no ETag, no measurement time	collecting since 8 Sept; the fifth state in production
Weather	RHMZ automatic stations (S01), METAR LYBE (S03), Meteostat 13274/13272 CC BY, Open-Meteo history	10 min – hourly	yes	probed
Water	RHMZ hydrology; DanubeHIS (ICPDR) with CC BY-NC-SA 4.0 and WaterML 2.0; groundwater Borča 9NP163 (S192, collected: 389 rows, 284 dates, 105 duplicate rows, 62 missing days)	daily / near-real-time	partly	one collected, others probed
Ground	Seismic SJ network (BEO since 1918, AVAS...; DOI, archive effectively empty); two Raspberry Shake citizen seismographs inside city territory via FDSN (found 8 Sept); INTERMAGNET GCK Grocka geomagnetic; EGMS ground motion	continuous	yes	probed

Electromagnetic	RATEL EMF drive tests and fixed stations (375 resources; one 225 MB archive with duplicates and 99999.99 unknowns)	irregular	yes	collected once; byte caps since
Energy	Elektro distribucija rolling outage schedule; EMS transparency (XML/CSV/JSON); ENTSO-E TP bidding zone RS; AERS plant tables	daily / hourly	yes	probed / new
Mobility	GTFS static (31 Oct 2025, SODL); ten street counters at the City Traffic Management Center (page only); JP Putevi Srbije AADT 2018–2024 (S57, collected: counters fell 317→227); MUP accidents monthly with geolocation (new)	annual / monthly	yes	one collected
Sky and light	Sentinel-2 and hourly Meteosat (CC BY 4.0) — the only lawful cameras; Sentinel-5P NO2; VIIRS nightlights; NASA FIRMS; AOB Urban Observatory SQM/TESS/all-sky (found 8 Sept)	daily / minutes	yes	probed / new
Radiation	EURDEP/REMon (SRBATOM 32 instruments since Feb 2024)	hourly	yes	new
Ionosphere / troposphere	DLR near-real-time TEC maps (Europe); EUREF EPN ZTD (BEOG inclusion unconfirmed)	15 min / weekly	yes	new
Prices	27 retail chains' price lists on data.gov.rs (weekly; per-store on every change from 1 Sept 2026 by regulation); JKP pijace "Barometar"; RGZ property price register	weekly	yes	new; registry S132 must move from no_coverage
Public health	GZJZ public fountains (32, monthly), epidemiological reports, duty pharmacies	monthly / daily	yes	new
Words	Eleven news/notice feeds (S68–S78), all permitted; Beoinfo RSS; GSP diversions; Službeni list (outside copyright)	continuous	publication only	six collecting since 8 Sept, headline-only

Memory	Internet Archive: 101 capture-days of parking back to 2014, 257 of beoeko back to 2005, 180 of river levels back to 2008; Digitalna NBS 1921 city plan (public domain); WSF Evolution annual built-up 1985–2015; GHSL; CORINE; Copernicus DEM	static	capture time	documented
Ground truth for space	GeoSrbija WMS/WFS (address register, NSTJ, census circles; DKP needs an agreement); H3 as the index spine (cell-area variation 0.61 % across Belgrade); Linked Places Format	continuous	—	PLACE_REGISTER adopted

6.2 The state layer, and the front door we had walked past

data.gov.rs runs udata with an API at `/api/1/`, 3 527 datasets, 233 organisations and 7 064 resources, and no robots.txt. BEOPS had taken one dataset from it before 6 September. Harvesting the catalogue and counting distinct resource hosts produced six new leads at zero network cost, and one count that became the project's most quotable sentence: **57 Serbian towns and municipalities publish open data; Novi Pazar publishes 67 datasets and the City of Belgrade publishes 7** — GTFS and one dataset from JKP Veterina Beograd; no parking, air, noise, waste, budget or GIS datasets from the City exist on the national portal, and the City has no portal of its own. Belgrade's 2026 budget (205.4 bn RSD) is published as gazette text; small municipalities publish theirs as XLSX.

The Serbian Open Data Licence 1.0 was added to the EU licence vocabulary on 17–18 June 2026, which makes a DCAT-AP export from BEOPS harvestable into data.europa.eu — the cheapest way for Belgrade to appear in any European table at all (§9).

6.3 The closed layer — what Belgrade has that Belgrade cannot read

What a city keeps closed is a fact about that city. The registry records 23 sources as genuinely unavailable, in five kinds that must never be spoken of as one: refused by name (Pleiades' robots.txt naming AI agents; Beobuild; Zoomer, Insajder, eKapija under Content-Signal; Politika, Vreme, Nedeljnik; three camera aggregators; transit.land; CORDIS bulk); open content behind a closed door (the ArcGIS server of the road authority answering `499 Token Required` — so the best traffic series in the country cannot be drawn; the Sava Commission; ICPDR TNMN behind an account); paid (the military-geographic institute's archive); request-required (aggregates from public utilities — letters drafted, none sent); and exists-unpublished.

The dark instruments, named: GTFS-Realtime fed to Google since February 2024 with no public URL; ~280 adaptive intersections; 35 noise points measured twice a year with 49 of 118 warning sirens dead as of 2022; district-heating substation telemetry; ~7 000 waterworks samples a year, programme described, results unpublished; the Institute of Physics' muon and radon stations with dead web pages; Plovput's 18 AIS base stations

publishing no positions; SOX Belgrade exchange traffic marked private; SRBATOM's own site offering PDFs to 2018 while feeding Europe hourly. And the absences confirmed so nobody repeats the search: no DNMS noise sensors, no Telraam, no Eco-Counter, no TomTom index page, no Waze partnership, no GBFS, no BirdWeather, no meteor cameras, no SatNOGS station in Belgrade, no AERONET site, no radiosonde since 1 April 2016.

7 • Findings so far

7.1 OBS-001 — the 10K race through the parking counters

Thirteen hourly slots were planned; the first three were lost because a schedule that lives inside an AI session is not a schedule (0 of 13 at 13:30 local; a durable Windows task then took the remaining ten). 270 displayed values from 27 lots, `observed_at` null throughout. Displayed free spaces summed 3 679 → 4 598 across the same 27 lots between the first and last reception — a difference between two receptions, not a measurement of the city. The pre-registered "race environment" group (Pinki, Belvil, Opština NBGD) produced the flattest lines in the set (+9, +7, +5); the movement was in the groups the protocol had called context (Kalemegdan 0→56, Donji grad 254→−209, Ada 440→954). The falsified pre-registration is reported as the result. Two headline artefacts were caught and corrected in the day: a −180 "largest drop" at VMA that reversed by +167 in the next slot (rule: in a series without measurement time, an excursion is evidence only if it persists through two consecutive samples), and a claim that Kalemegdan read zero in every sample when the raw 15:30 UTC receipt read 1. The Internet Archive later resolved what the live source could not: Kalemegdan and Kamenička, zero in every sample, read 101 and 57 in April 2025 — the field is not dead since forever.

7.2 The model source labels local time as UTC

On the first tick of the live collector (20:52:29 UTC, 8 Sept) SEPA's HVD API returned rows with `published_at_utc = 2026-09-08T22:50:17Z` and hourly intervals ending `22:00:00Z`. A publication two hours in the future is impossible; the field carries Central European Summer Time labelled Z. The source that ships its own epistemic status, aggregation period and legal basis as data — the best-behaved source in the registry — mislabels its clock. BEOPS keeps the label as received and adds `phenomenonTimeCorrected` as an *estimate* (minus two hours) with the reason, marked as estimated, to be re-checked after the clocks change on 25 October. This is the fourth clock in the project after measured, displayed and captured, and it belongs in the paper as a one-paragraph case: even a high-value-dataset API needs a receiver that can disagree with it.

7.3 False refusals, and the cost of "conservative"

The provenance index had recorded Kontur's H3 population dataset — CC BY 4.0, commercial use allowed — as having "said no", on the strength of `x-robots-tag: index, follow`. Conservative is not the same as true: a false refusal is a false statement about a publisher and silently removed the only open population layer from v1. The correction and its test are C-010; the closed-layer document of 6 September, which counted "10 refused", stays dated and unedited.

7.4 What the first unattended day shows

The collector's own report for 8 September (generated, not written) is the first output of the kind the project exists for: a quorum row per source, one cell per cadence slot, filled for a reception, hollow for silence, with the last reception's age. On day one it is mostly hollow, because the day started at 20:52 UTC; that is what it should look like, and a display that made it look better would be lying. Studio B's feed reset the connection (recorded as failed, no rows). Tanjug's Belgrade feed carries items back to July, so publication times spread over two months at first reception — a property of the feed, recorded as such.

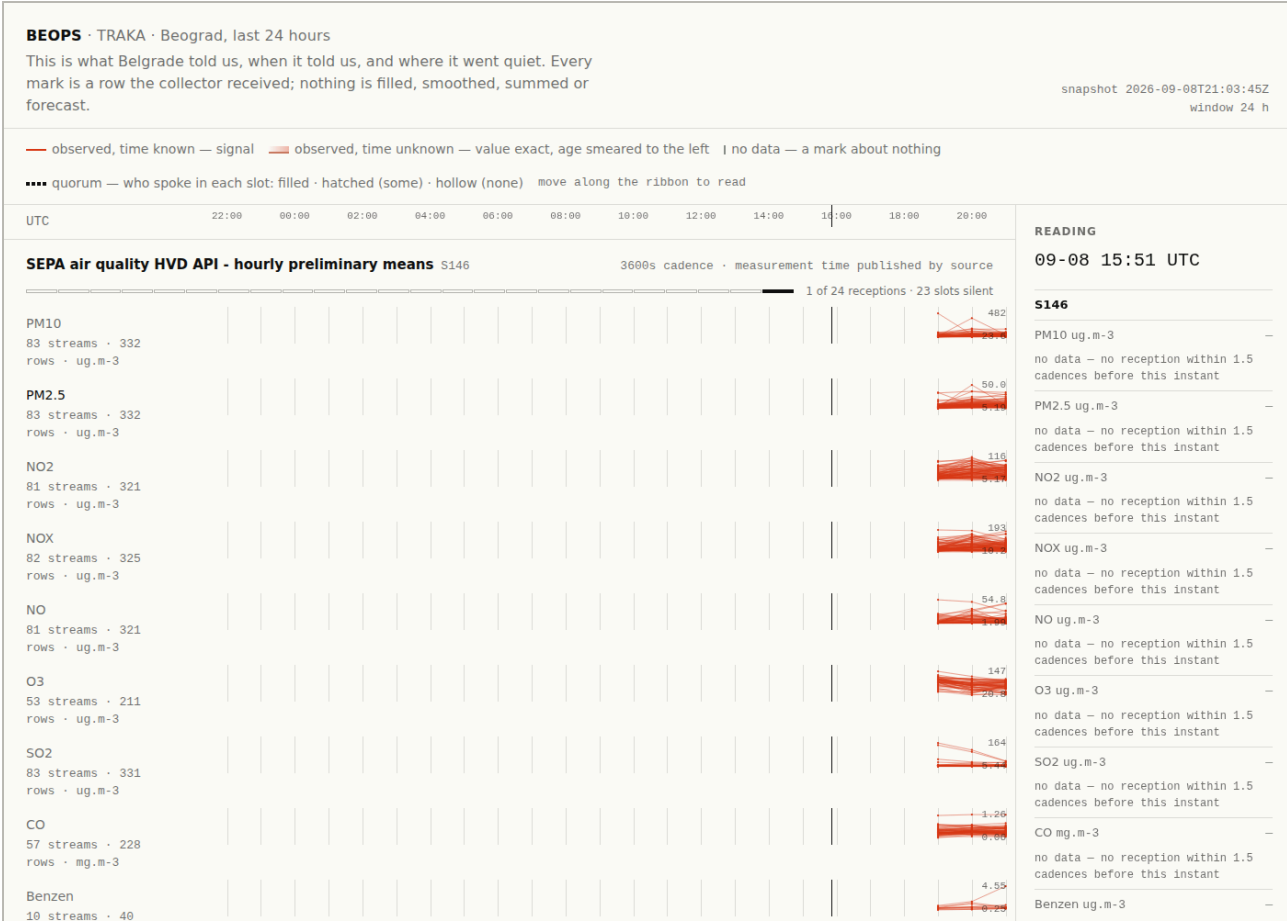
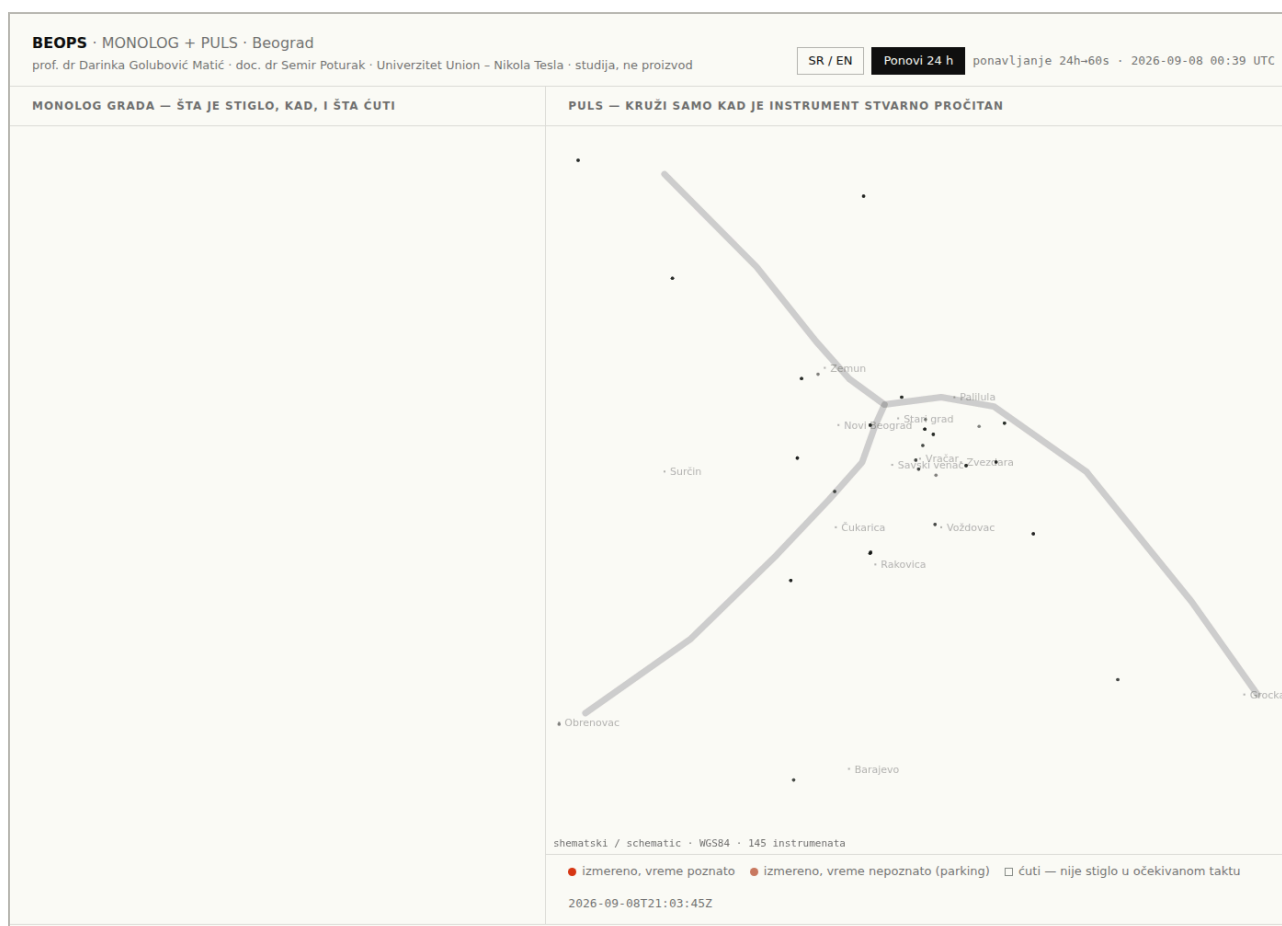


Figure 1 — TRAKA over the live snapshot of 8 September, 21:03 UTC: lanes per sense, marks only where rows exist, the reading column at the hairline stating who spoke. Most of the 24-hour ribbon is empty because collection began at 20:52 UTC; nothing is drawn where nothing arrived.



```
phenomenonTime (interval or instant, or null) · phenomenonTimeUnknown (boolean) ·  
phenomenonTimeReason (why) · phenomenonTimeCorrected {value, state:"estimated", why} ·  
resultTime · receivedTime (always) · resultQuality ∈ {unvalidated, source-preliminary,  
source-verified, missing} · permission_capture (id of the stored permission) · raw_sha256  
· dedupe_key · sourceClockNote
```

Absence is modelled at three levels: the row (`result:null`, quality missing), the receipt (state failed/unparsed/source_paused with the HTTP status), and the slot (a cadence slot with no receipt — computed, never stored, so it cannot be back-filled). Zone-level absence is shown as a quorum, never as a percentage. Every model output is a separate row type (`beops-derived-row/v1`, state estimated) with model id, prompt hash and organ version; a derived row can never be mistaken for an observation because it lives in a different file with a different schema.

What to borrow from the survivors (§9): Newcastle's Entity → Feed → Timeseries split with several series per feed so raw is never overwritten by derived; Amsterdam Schema's lifecycle fields (`status`, owner, licence, validity intervals); Aarhus's honesty about cadence ("~10 min, varies 2–30 depending on weather and season"); the UK sensor report's rule that quality is purpose-relative ("fit for city-wide trend, not for point compliance"); a one-file DCAT-AP catalogue so udata/CKAN can harvest; DuckDB CLI and Datasette as the read/publish layer over the JSON, no schema migration, no server as core.

9 • Comparison: observatories, platforms, and where Belgrade stands

The first generation of city dashboards is mostly dead or frozen, and the way they died is instructive. CASA's CityDashboard London (2012) now serves an expired certificate; Waag's Amsterdam City Dashboard carries a one-line epitaph; Sheffield's Urban Flows portal returns a page titled "Old Index"; Array of Things froze in 2020 and became an edge-AI testbed; Sentilo's code has not moved since December 2022; CityPulse died in 2019 but Aarhus at least labels its dataset "no longer updated, historical use only"; Every Noise stopped the day its data pipe was cut. Smart Columbus spent \$40 M and "was no longer used soon after demonstration funding ended in 2021" — no metadata standards, no federated provenance, a funding cliff (Wolf et al. 2025). The UK observatories survived through university support, modest ambition and standards.

What survives is a different genus: single-idea, relentlessly maintained, honest about sources. `earth.nullschool.net` prints the cadence of every source on one page and says its ocean data "is preliminary if fresher than two weeks and may change"; Blitzortung colours only by age ("white now, dark red 20 minutes ago") and disclaims itself; Raspberry Shake's helicorders are pure time-as-primary-object. Newcastle's Urban Observatory (4 000+ sensors, ~20 billion points, open API) is the closest analogue — and its quality flags are still "placeholder for future information", and no licence is visible on its entry pages. BEOPS has quality flags and permission per row from day one; it has one day of data.

Among Balkan peers, Zagreb (CKAN 2.10, 199 datasets, no air or parking) and Sofia (CKAN 2.11, 266 datasets, real-time GTFS) publish through their own portals; Belgrade

publishes seven datasets on the national one. Belgrade sits outside every comparative European city framework: not in Eurostat's Urban Audit (`urb_cpop1?cities=RS001C` returns nothing), not among the 112 Mission Cities (Sarajevo and Podgorica are), JRC's Urban Data Platform unverifiable (in maintenance on 8 September). The one table Belgrade is guaranteed to be in is the global GHS Urban Centre Database R2024A. That gap is the project's academic justification and its duty: a one-person observatory becomes citable by being harvestable.

10 • Position in the literature

Sixty-three works and standards were verified on 8 September 2026 on top of the project's eight dossiers (full list with reading levels in `research/_trail/SPOLJNI_PREGLED_2026-09-08_ISTRAZIVANJE.md`). The honest map of what is open and what is not:

Claim BEOPS could make	State of the literature (2024–2026)	Verdict
A per-observation semantics and rendering for <i>unknown measurement time</i>	OMS/ISO 19156:2023 makes phenomenonTime mandatory; STA 2.0 lets resultTime be omitted silently; OWL-Time has no "unknown" instant; no visualisation study exists (Wang et al. 2021 survey; Tennant & Randall 2025 scoping review has no such category)	Real gap. Frame as an OMS/SOSA extension profile plus the age-band rendering; test it (§12)
Absence as a spatially aggregated first-class record with a visual grammar	Missing-data visualisation is tabular/clinical (Alsufyani et al. 2024; Fernstad et al. 2025 "To Measure What Isn't There"; Ruddle 2026); IoT data-quality reviews treat gaps as imputation targets (Ferrer-Cid et al. 2025)	Real gap. The quorum bar is the candidate; cite the imputation mainstream as the position BEOPS rejects
Data age / staleness in a public display	Only vendor blogs; no controlled study found	Real gap; a user study is cheap and publishable on its own
Verification of sources discovered by AI agents	"Cited but Not Verified" (2026: links valid >94 %, facts 39–77 %) and CiteCheck (2026) exist for scholarly citations; nothing applies retrieval-grounded verification to web data sources (schema, licence, time semantics, robots signal); StreamReason-Bench (2026) shows no LLM exceeds 34 % on event-time semantics	Novel in application, not mechanism; state it that way
Serbia-specific sensing coverage	Last completeness evidence for SEPA is journalism from 2020 (50–80 % unusable); 2024–2025 low-cost-sensor papers publish no outage statistics; the 2026 survey of 99 urban digital twins has no Serbian or Balkan case; ODM 2025 says the portal hides unavailable datasets	A documented Belgrade map of absence would be the first
Forecast is not measurement	OMS <code>validTime</code> , DQV, DCAT provenance	Not a gap; BEOPS contributes enforcement and rendering only
Provenance for sensor data	SSN 2023 Edition, DCAT 3, PROV-O, time.IO/SaQC stack (Bumberger et al. 2025)	Standard practice; cite, do not claim

Critique of dashboards and twins	Saturated: Kitchin 2014 → Dawkins & Kitchin 2025 (twins for participatory steering), Borbach et al. 2025, Moroni 2025, Batty 2025, Azadi et al. 2025, Hiller et al. 2026	Position BEOPS as an <i>instance</i> of a "steering representation", not as a new critique
AI agents for dataset discovery	Two surveys (Li et al. 2025; Gan et al. 2026), a blackboard multi-agent architecture (Salemi et al. 2025), a user study (Kreutz et al. 2025: LLMs "a tool, not a partner")	Only the permission/time-verification angle is open
Legal TDM gap in Serbia	The fact is documentable; the analysis is covered (Quintais 2025; Senftleben 2026; Teilmann-Lock & Savin 2025; Kneschke v LAION, OLG Hamburg 10 Dec 2025; CJEU C-250/25 pending)	Document the fact and the collection policy built without an exception

Two papers change the paper: Longpre et al. 2024 ("Consent in Crisis": 45 % of tokens' domains restricted by terms within a year) grounds **permission** as a first-class attribute; Wolf et al. 2025 (why Smart Columbus died and UK observatories lived) is the benchmark for the sustainability section. The IETF AI-preferences vocabulary (draft-07, August 2026) carries an explicit *unknown* state — a parallel to BEOPS's permission field worth one sentence.

11 • The critiques we will get, and the honest answers

Critique (source)	Honest answer	Must not claim
"A city is not a computer" — dashboards flatten tacit and institutional knowledge (Mattern 2017)	BEOPS is an evidence ledger, not a model of the city; its unit is a source record with provenance and time semantics, and "missing" is a first-class record — it foregrounds what is <i>not</i> known	That BEOPS represents or understands Belgrade; that coverage is complete; the phrase "digital twin"
Real-time city / dashboard epistemology — indicators are performative, dashboards normalise a technocratic view (Kitchin 2014; Kitchin, Lauriault & McArdle 2015)	Adopt Kitchin's own remedies from the Dublin Dashboard: metadata, lineage, known unknowns; no ranking, no scoring; three timestamps so users see latency instead of a false "now"	"Real-time" as a headline — the parking source proved it cannot be honoured
Smart-city extractivism (Sadowski 2020; Greenfield 2013; Morozov)	Only what the city already publishes; no sensors, no cameras, no vendor lock-in, stdlib-only; any commercial layer sells access and service, never the evidence	"Smart city" branding; optimisation, control, efficiency gains
Data colonialism without persons (Couldry & Mejias 2019)	Zone/city level only; written refusal to derive person-level inferences; add a published "no ranking of zones" rule and aggregation thresholds	That aggregation alone removes harm; "hot spot" language
"Open data is not enough" / open-washing (Janssen et al. 2012)	The catalogue is an <i>audit of openness</i> : licence, machine-readability, timestamp presence; 7 vs 67 datasets; a source with no measurement time	That Belgrade "has open data" because portals exist

"So what?" — evidence without a decision-maker	Name first users (journalists, NGOs, FOI requesters, students) and one concrete question answered ("when was this figure measured?"); one pilot is a proof of method, not impact	Policy impact; any causal claim from ten hours of data
Western Balkans civic-tech fatigue, donor dependence	Near-zero continuation cost; a published sunset plan: open licence, public repository, evidence hashes; the graveyard in §9 acknowledged	Longevity; "permanent"
One-person sustainability, bus factor	Two authors, AI agents as instruments, everything reproducible from stored hashes; propose an external re-run of the catalogue by a student lab	Institutional backing beyond what exists
AI-generated research scepticism; hallucinated citations (Nature 1 Apr 2026; Retraction Watch 7 May 2026)	Every citation human-verified with DOI/URL and a reading level; every record hashed; the agents' role is discovery and cataloguing; a verification-ledger appendix	That the AI verified anything on its own
Journal AI-use policies (Elsevier: AI in the research process goes in Methods; COPE; Springer Nature)	The agent pipeline in Methods: models, prompts, versions, dates, human checkpoints; the standard declaration	Agents as co-authors; anything hidden
Reproducibility of LLM-driven collection (non-determinism even at temperature 0)	Reproducibility is of the ledger, not the discovery; discovery is a documented search procedure with unknown recall	That discovery is reproducible or exhaustive; that 189 is "all"
Legal risk of scraping; no TDM exception in Serbia	BEOPS relies on licence + captured permission (167 of 189), not on an exception — stronger than an EU TDM claim; 22 uncaptured are not collected	Kneschke v LAION as Serbian authority
robots.txt has no legal weight (Ryanair v PR Aviation; hiQ v LinkedIn; Kneschke: opt-out must be machine-interpretable)	Both robots.txt and human-readable terms are honoured and stored, satisfying the Hamburg standard and Ryanair's contract point	That robots.txt is binding in Serbia — it is evidence of good faith, not a licence
ZZPL — news feeds contain names	Headline, link and publication time only, city level, no profiling; research regime and Art. 88; a DPIA-lite appendix	"No personal data in a news feed"; say "no person-level records are retained or derived"
"Observatory" branding (UN-Habitat GUO/LUO; Newcastle's sensor network; URBEL's own GIS)	Define it in the first paragraph as an evidence catalogue with time semantics; invite URBEL as a source, not a competitor	UN-Habitat affiliation; official status

How comparable projects answered: Dublin published its own critique and added lineage; Array of Things published a privacy policy and a governance board; Newcastle wrote honestly about sensor failure in the wild; Helsinki adopted ethical principles for data and AI; Amsterdam, Helsinki and seven cities set an algorithm-register standard (January 2023). BEOPS's cheapest direct answer to three critiques at once is already built: `research/ORGANS.json` is an algorithm-register entry per model.

12 • Design

12.1 The sentence and the alphabet

Every artefact exists to say one sentence: **This is what Belgrade told us, when it told us, and where it went quiet.** The material grammar that follows has two axes — *did a source say it?* (stroke) and *do we know when?* (ground) — so the five states differ in two physical properties before colour is applied. Magnitude is always position or length; colour is redundant, never sole; absence is a positive mark; a forecast never has an outer edge; a specific uncertainty ("age $\leq$ 61 min since reception") costs no trust while a vague badge ("quality: unclear") does (the 5 780-participant finding), so there is no confidence badge anywhere.

12.2 The compositions

TRAKA (the ribbon) — time is the primary object: 24 hours across the screen, one lane per sense, a hairline that reads every lane at once. Built on 8 September over real rows (`traka-live.html`, Canvas, zero dependencies): lanes per parameter with stations as faint traces, a quorum row per source, lanes that expand into stations, and a reading column stating "n of m spoke", "age $\leq$ x min, measurement time unknown", or "no data — no reception within 1.5 cadences". **KARTON** (the register) — the instrument is the primary object: a card that does not change shape when the data is bad. **PRESEK** (the section) — the city as an architectural drawing, the front door and the argument against a 3D twin. **MONOLOG + PULS** — added on 8 September at the authors' request: the city's monologue as the main interface, bilingual, one sentence per reception built deterministically from receipts ("SEPA published hourly means for 32 Belgrade stations — 2 477 rows; hour as the source labels it 21:00–22:00 UTC; our estimate of true UTC 19:00–20:00"), silence as a sentence ("Studio B is silent: 12 of 48 expected slots without a reception"), the organ's silence as a sentence, and beside it a schematic map whose points are instruments at their real coordinates and whose rings appear only when an instrument was actually read — replayable at 24 h $\rightarrow$ 60 s, or live.

12.3 Ten decisions, each with its precedent

1. Every card carries a nullschool-style provenance footer, always visible: source, cadence, last received, measurement time known / inferred / unknown.
2. Chroma is purchased with certainty (value-suppressing uncertainty palettes, Correll, Moritz & Heer 2018): saturation only where the measurement time is known and fresh.
3. Age is the only thing colour-ramped on the map (Blitzortung: white now, dark two hours ago).
4. Absence is hatch plus a one-word reason — dead, opted out, never — because "emptiness plus explanation" wins on decision confidence (Alsufyani et al. 2024); grey-for-no-data is a mass-market convention (Statuspage) that needs no legend.
5. A card never changes shape ("technology should work even when it fails", calm technology).
6. TRAKA draws 100+ zone rows as horizon bands on one shared Canvas on one frame clock, with a hidden DOM table as the accessible twin (Heer, Kong & Agrawala 2009; Canvas vs SVG guidance 2026).

- 7. Forecasts and estimates are structural, dashed or dotted, drawn *behind* measurement (Peak Spotting's separation of predictions from bookings); the Wind Map's forecast-drawn-as-live is explicitly refused.
- 8. Quorum cells are sized by contributing sensor count, so 3 of 12 and 30 of 120 look different at the same ratio.
- 9. Revisions are visible, never silent: a corrected cell keeps an under-tick and a "revised" tag, because trust dies from being slightly wrong too often.
- 0. The organs narrate arrivals, not values: a tick per reception, silence for absence (Listen to Wikipedia); intrinsic cues — blur, texture — for unknown time (Altimir 2026), never a lava lamp with hidden sources.

12.4 Systems

Field warm off-white #FBFAF7 / ink #111311; signal #D93A16 for observation only; the same hue at ~45 % chroma for untimed; one cool grey for state; two type families — a humanist sans for reading and a tabular monospace for every number, timestamp, hash and id; base unit 4 px; no fills, gradients, shadows or glow; the only animation about the world is a mark drawing itself in when a sample lands; prefers-reduced-motion honoured; contrast 4.5:1 on both themes; narrow screens rotate the ribbon so time runs vertically. Bilingual by construction: every user-facing string exists in Serbian and English, the language toggle is a first-class control, and the "How this works" section with links to the repository, the rules, the permission index and the organ register is part of the page, not a footer.

12.5 The cheapest test that could break this

A between-subjects study with three conditions — value only; value with age band; value with a textual "last updated" — measuring how often participants act on a stale parking figure. It is cheap, it uses real BEOPS samples, and its result is publishable whether it confirms or breaks the age band. It is also the test for literature gap 3 (\$10).

13 • The organ of mind

The authors' vision is intelligence through a complex organ of mind built from many small local models, where the main interface is the city's continuous monologue. The doctrine that keeps it honest is one sentence: **an organ derives; it never measures**. Every organ output is state "estimated", drawn structurally, stored in its own file with model id, prompt hash and organ version, marked `ai_generated=true`, and traceable to one input row. An organ is allowed to be silent, and its silence is a receipt.

Organ	Job	Model class (present on the PC or in the Svemir registry)	State
news-sorter	Sort headlines into a fixed topic list, say whether Belgrade, propose ≤ 3 zones from the gazetteer keeping every candidate with its score	local instruction models: qwen2.5:3b, qwen3.5:0.8b, lfm2.5:1.2b, llama3.2:1b (cloud-hosted names are skipped by rule — no headline leaves the machine)	built, tested, first run 9 Sept

series-watch	Flag unusual movements in measured series as estimates with an expected range; never fill a gap	ibm-granite TTM-r3 (1.4 M params), chronos-bolt-tiny, MOMENT-1-small; onnxruntime already installed	specified; needs seven days of rows first
daily-voice	The sentence of the day, SR and EN, from status and derived rows, labelled model-written, accepted or struck by the editor of record before it is shown; a test compares every number in the text with its sources	local instruction model	specified
twin-ear	Near-duplicate and cluster detection across feeds so one event is one sentence	djovak/embedic-small (Serbian e5 fine-tune), multilingual-e5-small ONNX	candidate in the registry
name-reader	Serbian NER for institutions and places in headlines (ekavica, administrative register)	ICEF-NLP bcms-bertic legal NER; GLiNER multi	candidate
claim-gate	A model-written sentence enters the monologue only if a document supports it	MiniCheck-DeBERTa-v3	candidate
guard	Prompt-injection screen on any text that comes from the web before an organ reads it	Llama-Prompt-Guard-2-86M	candidate; closes a real hole
eye (later)	Cloud fraction from Meteosat and the AOB all-sky camera; never a street camera	small vision models; PidNet 8 MB	research direction

Orchestration borrows Svemir's frequencies — reflex, breath (30 s), beat (5 min), waking (on event), sleep (daily), silence (never, on purpose) — with deterministic spacing so organs never all fire in the same beat, and the ladder reflex → organ → orchestrator → large model → human, where whatever a large model did three times the same way becomes an organ. The measured reality (Svemir, 5 September) that bounds the vision: inference is not bit-reproducible under load, so no model can be its own record of truth; 60–80 % of behaviour can be models, the 20–40 % of code is exactly the part that breaks everything if wrong; on this hardware the feasible shape is N llama-server processes sharing one GGUF base with small adapters. What no one has built and the literature does not have: re-performance in the shadow — checking from one's own logs whether another policy would have been better. That is the contribution the orchestration layer could make, later.

Law: the EU AI Act's Article 50 (transparency) applies from 2 August 2026 and would apply to BEOPS only if offered to EU users, but it is the right design target anyway: users told they interact with an AI (any Q&A layer), generated text machine-marked, and public-interest text disclosed as AI-generated *unless* it has passed human editorial control with a named person responsible — which is exactly the editor-of-record path in the organ register. Serbia's draft AI law (June 2026, four risk tiers, adoption expected end-2026) would place a public evidence dashboard in the minimal or medium tier with mirrored transparency duties. Practical obligations: label every model-generated sentence, keep a

human editor of record, publish an algorithm-register entry per organ, keep the evidence ledger separate from the generation layer so labelling is trivial.

14 • The public site, the domain, and open source

14.1 beops.rs

Checked on 8 September 2026 against the registry's own WHOIS server (`whois.rnids.rs`, port 43): **beops.rs is not registered** (%ERROR:103: Domain is not registered). Registration of a .rs domain goes only through an RNIDS-accredited registrar (*ovlašćeni registar*); the registry publishes the list and does not sell directly. The one registrar price page that answered a direct check that night was Plus (`plus.rs`): **.rs 1 999 RSD per year**, .co.rs 900, .in.rs 600 (prices as displayed, VAT status not stated on the page); Loopia, Ninet, EUnet, SBB, Beotel, Hostgate and mCloud pages were unreachable or moved in the same check, so their prices are unverified. The cheapest honest path: (1) pick an accredited registrar from RNIDS's list, compare the first-year and the *renewal* price (many discount the first year and recover it on renewal), and register for two or three years at once if the multi-year price is lower; (2) do not buy hosting from the registrar — the site is a static export plus one JSON snapshot and can live on any static host, including GitHub Pages for zero cost, with the PC pushing the snapshot; (3) register the Cyrillic .cp6 only if the project will actually publish in Cyrillic (it publishes ekavica Latin and English). Expect roughly 1 300–2 000 RSD per year all-in for the .rs alone; anything much cheaper is a first-year promotion.

14.2 Hosting shape

The observatory must never depend on the site: the collector runs on the university PC and writes files; the site is a reader. Version 1 is a static export — the two studies and the snapshot — pushed every few minutes to a static host, which is enough for the monologue and the pulse to be live within a five-minute lag and costs nothing. Version 2 adds a small read-only server (the existing `server.js` pattern, loopback-hardened) behind a reverse proxy on a cheap VPS or the university's own machine, serving the DCAT-AP catalogue, the Frictionless packages and a Datasette view. Nothing is written from the web; there is no user account; the only inbound is a request for a page.

14.3 Bilingual by construction

Every user-facing string, every sentence of the monologue and every organ output exists in Serbian (ekavica, Latin) and English; the language toggle is a first-class control; the documents that go public are English with Serbian working notes kept as such. The "How this works" section is part of the page: what runs where, the three timestamps, what null means, links to the repository, the rules, the permission index and the organ register — in both languages, with the authors named.

14.4 Open source

The repository is published as **github.com/3esign/beops** under MIT for code and CC BY 4.0 for documents and registries (`LICENSE.md`), through `tools/publish_github.ps1`: an export of the current tree minus the captured evidence (585 MB of third-party pages kept locally

as proof, with every hash listed in the index so a reviewer can request any capture by id), minus scratch, archive, runtime and live data, as one commit per publish naming the local commit. Local history is never rewritten; the 141 MB evidence blob that GitHub refuses stays where it is. Values keep their source's licence; headlines are never republished as a corpus; reservations against training are honoured and carried in the licence text.

15 • Commercial framing and funding

The pattern across every comparable product is the same: the observatory itself is never the product. Windy sells an ad-free premium (about £23 a year after a 24 % rise in 2025); Flightradar24 earns SEK 419 M on freemium plus B2B on top of 40 000 volunteer receivers; Telraam sells per-device dashboards to municipalities (€15–25 per device per month, a ten-device network with a year of service €4 600); Vianova and Replica sell analytics to agencies with no public prices; PurpleAir metered its API in 2023 and made its own sensor owners furious at paying for their own data; Newcastle offers "bespoke monitoring and analytical services" as the university variant; EU data-space money (DS4SSCC, €15 M in cascade funding) pays for pilots.

The position consistent with "evidence-first": the ledger stays open (CC BY / ODbL), feeders are credited on screen and never metered, and what is sold is *freshness, uptime and interpretation* — API tiers with SLAs, alerting, institutional subscriptions for municipalities, real-estate analytics, insurers and media, per-zone dashboards at Telraam-like price points, and the organ layer as a service where "no citizen query leaves Serbia" is the selling point (ZZPL territorial scope, no third-party training on city data, auditability through the published register). Public-sector sovereignty procurement is a live trend (Mecklenburg-Vorpommern moved 50 000 staff to EU open source in July 2026; Schleswig-Holstein's OGP commitment; Vienna's 2025 open-source report), though no verified case of a municipality running an on-prem LLM in production was found — claim the trend, not a case.

Path	Realism for a two-author university project	Notes (verified 8 Sept)
University spin-off under the Law on Innovation Activity (2021)	High; prerequisite for most of the rest	Registration in the national innovation-system registry unlocks incentives; the university's IP rules decide the split
Innovation Fund Serbia — Mini Grants	High after the spin-off	Up to €120 000, 70 % grant, micro/small company ≤10 years, Serbian majority; last call closed April 2024, next not open at the check
Institutional subscriptions (service, not data)	Medium; needs a year of series	The only model that does not betray the ledger
Interreg Danube Region	Medium, as a partner	Third call closed 15 Dec 2025; SO1.2 AI skills, SO4.2 governance fit; consortium required
Horizon Europe 2026–27 (Cities Mission, Cluster 5)	Low as coordinator, medium as partner	Serbia associated; WP adopted
Digital Europe data spaces, EIC Pathfinder	Low for this project alone	The authors' separate EIC plan exists; not this document's subject

Data licensing to media	Marketing, not revenue	Small market
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16 • The paper

The conference "Creating sustainable commUNiTy" (University Union – Nikola Tesla, on-line, 25 September 2026) had its full-manuscript deadline on 20 August; it has passed, the late-submission enquiry is drafted and unsent, and its sending is the authors' decision. Independently of it, one contribution per paper, from a series rather than from a catalogue:

Paper	Contribution	Venue (verified dates/APC)
1 • The audit of openness	189 records coded, 167 permissions captured, the closed layer in five kinds, 7 vs 67, the SEPA clock label; the collection policy built without a TDM exception	<i>Data & Policy</i> (Cambridge; OA, APC waived when unfunded; post-conference route deadline 31 Mar 2027); <i>Places and Technologies 2027</i> (Faculty of Architecture, Belgrade; the home audience where critique 15 will surface; 2027 call expected Feb-Mar)
2 • Unknown time, counted absence	The OMS/SOSA extension profile, the age band, the quorum bar, and the user study of §12.5	AGILE 2027, Lund (AGILE-GISS, OA, no APC; deadline typically February); <i>Computational Urban Science</i> (APC £1 240, waivers case by case)
3 • The monologue and the organs	A public evidence interface driven by small local models under Art. 50 labelling with an editor of record, and the re-performance question	CUPUM 2027, Sydney (papers 12 Feb 2027); <i>Journal of Urban Technology</i> for the critical framing; MDPI <i>Smart Cities</i> only with eyes open (CHF 2 000 APC, reputational risk with this audience)

Claims ledger for any of them, from the existing methods audit: the register contains 189 records (verified snapshot — report unit and date); the register is the complete IoT infrastructure of Belgrade (unsupported — exclude); the pilot is a ten-reception descriptive observation with missing source clocks (verified); the parking observations establish the effect of the race (unsupported — exclude); a functioning city intelligence has been demonstrated (unsupported — use "proposed framework"); AI assisted extraction and synthesis under human verification (supported — describe in Methods with models, prompts, versions, dates, checkpoints). Disclosure: the agent pipeline in Methods plus the standard AI-use declaration; no AI co-authorship; a verification-ledger appendix.

17 • Risks and open decisions

Risk	Now	Decision or action
One disk	All history on D:; GitHub refuses the 141 MB blob; no remote	Authors: run <code>publish_github.ps1</code> (public export) <i>and</i> add a bare remote on the laptop or an external disk for the full history; move <code>research/evidence/</code> to a vault with the existing manifests

Deployment lies	C-012: a transfer reported written, content unchanged, a failing test did not stop the next command	Hash check on the body before the first dependent command; && after tests — done
Model daemon	Ollama serve was not running until 8 Sept 23:15; cloud-hosted model names appear in the local list	Local-only policy in code and register — done; a scheduled check that the daemon answers, like the collector's own
The source's clock	SEPA labels CEST as Z	Kept as received plus a marked estimate; re-check after 25 Oct; write to SEPA only after a week of evidence, as a courtesy, not a request
Legal	No TDM exception; headlines; robots.txt weight; ZZPL article numbers to confirm	Six questions already marked for a lawyer; nothing published with a name in it; the gazette corpus as the safe expansion
Scope creep	Twelve sources configured after one day; 37 + 60 candidates found	Seven unattended days before any new source; candidates enter only through a permission capture
Sustainability	Two authors, one PC	Zero-dependency stack, open licence, sunset plan in the paper, a student re-run of the catalogue
Conference	Deadline passed; nothing sent	Authors' decision; the shortest channel is the Scientific Committee

18 • Roadmap

When	What	Done when
Week 1 (to 15 Sept)	Seven unattended days; daily generated report; organ news-sorter on local model over the backlog; hash-checked deployments; backbone off the disk	Seven REPORT_*.md files; a remote that holds the history
Week 2	Registry 2.0 fields (licence id, cadence nominal/observed, phenomenon-time boolean, lifecycle, owner, H3 cells) by migration tool with a no-loss test; S132/S62 statuses re-captured; permission captures for the 25 undocumented and for the first ten of the 37 new sources with sub-daily cadence (Sensor.Community T/RH/P, DanubeHIS, ENTSO-E RS, EURDEP, CAMS, VIIRS, MUP accidents, GeoSrbija WFS)	Registry validates; index shows fewer than 10 undocumented
Week 3	Site v1: static export of MONOLOG+PULS and TRAKA with the "How it works" section, bilingual, on beops.rs via a static host; DCAT-AP catalogue file; Frictionless packages for the three live sources	The page shows last night's receptions with a five-minute lag
Week 4	Organs 2 and 3 (series-watch on seven days of rows; daily-voice with editor acceptance); the guard organ in front of every web text	A model-written sentence of the day, accepted or struck, every morning

Weeks 5–8	Paper 1 drafted from the series and the audit; the §12.5 user study prepared; Places and Technologies abstract when the call opens	Manuscript with claims ledger and disclosure
Quarter 2	PRESEK; the gazette corpus; a second city's <i>method</i> transfer (not a second city in BEOPS); the spin-off question	—

Honest verdict on this document. Checked: every count in §4 on files listed on 8–9 September; every URL behind §6, §9, §10, §11, §15 opened by a research mind that night, with its own verified/partial/unverified marks preserved in the trail file; the collector, the scheduler, the organ run and the WHOIS answer on the PC itself. Concluded: BEOPS is a defensible audit of openness with a working collector and a design doctrine that the literature does not yet have; it is not yet an observatory with series, and will not be for another week. Not checked: the text of the Serbian draft copyright law and the draft AI law; the AG opinion in C-250/25; registrar prices other than Plus; Urban Atlas coverage of Belgrade; whether the SEPA clock label is the same in winter; the contents of the 585 MB of evidence file by file.

Appendix A • New sources found on 8 September (37)

#	Source • owner	Contents • cadence	Access • licence	Time?	Status
1	Traffic accidents by police directorate and municipality • MUP	Geolocated accidents with date/time, type, casualties; 2026 to 31 Jul; monthly	XLSX on data.gov.rs, udata API; SODL	yes	Verified
2	rsnezgode black-spot map • ODI	~150 high-risk 200 m clusters from MUP data	Web map	inherited	Verified
3	Retail price lists • 27 chains via Ministry of Trade	Per-product prices, promo dates; weekly; per-store on every change from 1 Sept 2026	CSV/XLSX; SODL	yes	Verified
4	Industrial air emissions register • SEPA	Facility, pollutant, annual quantity 2010–2024	CSV; SODL	year	Verified
5	Waste registers NRIZ/LRIZ • SEPA	Type, quantity, location 2010–2024	CSV; SODL	year	Verified
6	GeoSrbija OGC services • RGZ	Address register, cadastre (DKP), NSTJ, census circles, TK250; continuous	WMS/WFS; no open licence stated; DKP needs an agreement	no	Verified
7	URBEL maps • Urbanistički zavod	GUP 2041, plan boundaries, geology	Viewer, DWF; "informational" terms	no	Verified
8	GIS Beoland • Direkcija za građevinsko zemljište	12 layers incl. landslide cadastre, land value zones	Web GIS; "informational"	no	Verified
9	DanubeHIS • ICPDR (RHMZ data)	Water level/temperature at Beograd, Zemun, Pančevo; 1–2 h old	WaterML 2.0/CSV for registered users; CC BY-NC-SA 4.0	yes	Verified
10	vodostaj.rs • private	RHMZ levels mirrored, 30 min	Web; no licence	yes	Verified
11	EMS Transparency • Elektromreža Srbije	System load, NTC, auctions; hourly	XML/CSV/JSON; disclosure policy, no reuse licence	yes	Verified
12	ENTSO-E TP zone RS	Load, generation, flows, prices; hourly	REST with free token	yes	Partial
13	Sensor.Community 25 km area	~34 PM sensors + ~49 T/RH/P; 2.5 min	JSON, no key; ODbL (prior knowledge)	yes	Verified; now collected

14	Klimerko.org	Community PM/T/RH map	Map only; API undocumented	yes	Partial
15	PurpleAir Belgrade	Sensors listed	REST with key, points-based	yes	Partial
16	WAQI feed of SEPA network	121 stations, ~12 Belgrade; hourly	JSON with token; "unvalidated, may be amended"	yes	Verified
17	EEA Air Quality Download Service	RS in country list; E2a hourly from 2023, E1a validated 2013–2022; Parquet per station	REST; EEA reuse	yes	Verified
18	CAMS European AQ forecasts · Copernicus	0.1° hourly analyses/forecasts	CDS API; CC BY	yes	Verified
19	EURDEP/REMon gamma · JRC + SRBATOM	32 instruments since Feb 2024; hourly	Map; files by agreement	yes	Verified
20	Meteostat 13274/13272	Hourly/daily station history	JSON, CSV; CC BY 4.0	yes	Verified
21	Open-Meteo historical	ERA5-Land hourly since 1940	JSON; free non-commercial	yes	Verified
22	NASA FIRMS	VIIRS/MODIS hotspots; ~3 h	CSV with key; open	yes	Verified
23	Sentinel-5P via Copernicus Data Space	NO2, SO2, CO, O3; daily	STAC/OData/S3; open	yes	Verified
24	VIIRS Black Marble	500 m nightlights daily/monthly	HDF5; open	yes	Partial
25	Open Charge Map Serbia	Charger locations and status type	JSON with key; CC BY-SA (prior)	partial	Partial
26	RGZ property price register	Every notarised transaction; 1–3 month lag	Web search, tabular export; no licence	yes	Verified (secondary)
27	Beogradske pijace Barometar	14 staples × 14 markets; weekly	HTML	yes	Verified
28	GZJZ public fountains	32 fountains; monthly microbiology	HTML/PDF	yes	Verified
29	GZJZ epidemiological situation	Weekly reports	HTML/PDF	yes	Verified
30	Duty health institutions · Grad Beograd	Daily roster	HTML + PDF	yes	Verified
31	Beoinfo RSS · Grad Beograd	City news, works, outages	RSS	yes	Partial

32	GSP regime changes	Line diversions	HTML	yes	Verified
33	Belgrade airport WebTrak, noise maps · VINCI	Aircraft tracks; strategic noise maps	Web app; PDFs; vendor terms	yes	Verified (existence)
34	AERS generation tables	Per-plant annual data	XLS	year	Verified
35	Poverenik open data portal	FOI statistics by authority	portal	?	Partial
36	Open Data Hub Serbia	Policy news	HTML	—	Verified
37	Copernicus Urban Atlas 2018	27-class land use, 0.25 ha	Download/WMS; open	no	Partial — Belgrade inclusion unconfirmed; registry S62 says no_coverage; re-check

Appendix B • Devices round 5 (condensed)

Open and machine-readable now: two Raspberry Shake citizen seismographs inside city territory (AM.RCE5A since Oct 2022, AM.RE172 since Feb 2023) via FDSN station/datasetselect, miniSEED; ~49 Sensor.Community T/RH/P sensors (a new variable class); DLR near-real-time TEC maps (Europe, 15 min); EUREF EPN troposphere product (BEOG inclusion unconfirmed); Windy Webcams API as the lawful route to twelve Naxi-origin Belgrade cameras (attribution, original size, link back — image tokens expire in 10 min on the free tier); Ookla open data tiles (CC BY-NC-SA 4.0, quarterly); Cloudflare Radar Serbia (country level); Open Charge Map; STARS4ALL TESS-W photometer stars1200 at the Astronomical Observatory tower (Grafana dashboards).

Open page, scrape-only: the Astronomical Observatory's UrbObsBel suite (SQM-LE at the tower and at ADRB, all-sky camera, data repository) under Science Fund project 6775; Vidojevica station as a dark-sky reference; VojvodinaMeteo's index of six Belgrade points including four live Weather Underground stations (Kotež, Batajnica, Borča north and south); Sava water temperature via vodostaj.rs; Beogradske elektrane system-state posts; LiveATC Belgrade Tower audio (not for redistribution); D-ATIS text; the AGROS rover monitor; the Belgrade TTN community (four LoRaWAN gateways); iNaturalist place 14703; SRBATOM's early-warning page.

Exists but closed: EUREF BEOG real-time (registration); AGROS (registered legal entities); the Institute of Physics' muon and radon stations (pages dead since 2017); Plovput's AIS base stations (no positions); airport noise terminals; municipal noise campaigns; Naxi cameras by terms; SOX exchange traffic; charge&GO live map; Weather Underground API (station owners only).

Confirmed absent or not found: DNMS noise sensors within 25 km; Telraam; Eco-Counter; TomTom index page for Belgrade; Waze partner; GBFS (partial read of the systems list — re-check rows R-S); BirdWeather; AllSky7 and Global Meteor Network in

Serbia; SatNOGS in Belgrade (one station in Novi Sad, last seen 2022); KiwiSDR/WebSDR, APRS iGates, Netatmo, Weathercloud (unverified, blocked); institute weather stations, phenology, zoo; "smart city" pilot dashboards; street-lighting, gas and waterworks telemetry; per-city ADS-B feeder counts; NoiseCapture data for Belgrade.

Appendix C • Row and receipt schemas

```
beops-observation-row/v1 – sid · datastream · station_id · station_name · parameter ·
lat · lon · result (number | string for text rows | null) · unit · phenomenonTime
({start,end} | instant | null) · phenomenonTimeUnknown · phenomenonTimeReason ·
phenomenonTimeCorrected {start,end,state:"estimated",why} · resultTime ·
resultTimeCorrected · receivedTime · resultQuality (unvalidated | source-preliminary |
source-verified | missing) · aggregation · source_status · sourceClockNote ·
spatial_binding · permission_capture · raw_sha256 · dedupe_key · kind (text for
headlines) · link
```

```
beops-live-receipt/v1 – sid · name · url · attempted_at · permission_capture · state
(captured | failed | unparsed | not_permitted | source_paused | already_recorded |
claimed_unfinished) · http_status · transport · headers_of_interest · raw_file |
raw_not_stored · raw_sha256 · raw_bytes · rows · rows_new · rows_missing · rows_file ·
error · paused_source · completed_at
```

```
beops-derived-row/v1 – state:"estimated" · organ · organ_version · model ·
prompt_sha256 · ai_generated:true · input_sid · input_key · input_headline ·
input_resultTime · derivedTime · category · belgrade · zones [{name|null, score,
rejected?, why?}] · binding:"inferred_from_content" · event_time_text · note
```

```
beops-organ-receipt/v1 – organ · organ_version · at · waiting · state (nothing_to_do |
organ_silent | derived | organ_failed) · derived · model · calls · errors · reason ·
output
```

Appendix D • Corrections C-001 to C-012

Id	What the system said	What was true	Rule that followed
C-001	Seven captures reported allowed_for_us = true	Every request had failed on TLS; None is not False evaluated to True	Unknown and permitted are never the same value; three-state verdicts
C-002	Ids allocated to sources already registered	Duplicates S134/S136/S137 of S57/S95/S04	The tool refuses unregistered ids and duplicate hosts
C-003	Data taken before permission was captured	Wikidata/Overpass queried first	Evidence before collector; held as needs_decision
C-004	The verdict engine over-permitted, by standard	Longest-match vs first-match disagreement between parsers	Two engines, take the stricter

C-005	"Take the stricter" turned one tool's bug into a refusal	A parser bug read as a refusal	Engines disagreeing is recorded, not silently resolved
C-006	A valid certificate reported as unverifiable	The Python on the PC had no CA bundle; the pilot's transport was touched	Transport through the OS store; no unverified mode
C-007	An EMF file associated with a NetTest capture	Wrong host behind a source-id gate	Exact-route enforcement; additive correction
C-008	Access-check wording overstated legal clearance	An access signal is not a licence	"Reuse remains source-specific" in the index
C-009	A captured 403 treated as access/licence text	A 401-byte error page stored as terms	Held offline; no alternate downloader
C-010	Three permitted sources listed under "said no"	index, follow and noindex are not refusals	Only noai/noimageai, TDM-Reservation: 1, Content-Usage ai=n/tdm=n refuse reading
C-011	The pilot task "self-deleting"	cmd IF GTR saturates at 32 bits; 69 empty ticks	A schedule's end is verified at the scheduler
C-012	The organ ran local-only	It picked a cloud-hosted model; the file on the body was old; a failing test did not stop the run	Hash on the body before the first dependent command; && after tests

Appendix E • Selected references

Full annotated lists with reading levels are in the repository (research/04-bibliography/ and research/_trail/SPOLJNI_PREGLED_2026-09-08_ISTRAZIVANJE.md). Standards: OGC SensorThings API 1.1 (18-088); OGC 20-082r4 / ISO 19156:2023 Observations, Measurements and Samples; OGC SensorThings 2.0 draft (23-019, 2025-26); W3C SSN 2023 Edition (WD 2026); W3C DCAT 3 (2024); DCAT-AP 3.0.1 (2025); W3C PROV-O (2013); W3C DQV; W3C OWL-Time (CR 2022); Frictionless Data Package v2 (2024); QUDT; IETF draft-ietf-aipref-vocab-07 (2026); Cloudflare Content Signals Policy (2025). Literature: Alsufyani, Forshaw & Fernstad 2024 (missing-data visualisation survey); Fernstad et al. 2025 (To Measure What Isn't There); Ruddle 2026 (PLOS One); Tennant & Randall 2025 (IEEE TVCG); Leffrang & Müller 2025 (J. Forecasting); Wang et al. 2021 (J. Visualization); Correll, Moritz & Heer 2018 (VSUP); Heer, Kong & Agrawala 2009 (horizon graphs); Altimir 2026 (Frontiers); Bumberger et al. 2025 (SoftwareX); Ferrer-Cid et al. 2025 (JNCA); Akhtar et al. 2024 (Croissant, NeurIPS); Li et al. 2025 and Gan et al. 2026 (dataset-search surveys); Salemi et al. 2025 (blackboard agents); Kreutz, Perry & Friedrich 2025; Onweller et al. 2026 (Cited but Not Verified); Khajavi et al. 2026 (CiteCheck); Wang, Zheng & Zhang 2026 (StreamReason-Bench); Longpre et al. 2024 (Consent in Crisis); Akidau et al. 2021 (watermarks, PVLDB); Dawkins & Kitchin 2025; Borbach, Chun & Thielmann 2025; Moroni 2025; Batty 2025; Azadi et al. 2025; Hiller et al. 2026; Wolf et al. 2025 (Frontiers in Sustainable Cities); James et al. 2022 (UK Urban Observatories Sensor Report); Stojanović et al. 2024 (Sensors); Ramadani et al. 2025 (Atmosphere); Arnaut et al. 2025 (Urban Science); data.europa.eu ODM 2025 Serbia; Mattern 2017; Kitchin 2014; Kitchin, Lauriault & McArdle 2015; Janssen, Charalabidis & Zuiderwijk 2012; Couldry & Mejias 2019; Sadowski 2020; Greenfield 2013; Quintais 2025; Senftleben 2026; Teilmann-Lock & Savin 2025; OLG Hamburg, Kneschke v LAION, 10 Dec 2025; CJEU C-250/25 (pending); Brown et al. 2024 (web scraping for research); Bogataj Jančič 2023 (Slovenian TDM). Belgrade primary sources are cited by source id in SOURCE_REGISTRY.json.

Appendix F • Glossary • Rečnik

Term	Srpski	Meaning here
observation	opažanje	a value a source reported, with its provenance
phenomenonTime	vreme merenja	when the measured thing happened, as the source says; may be unknown
resultTime	vreme objave	when the source produced or published the result
receivedTime	vreme prijema	when BEOPS received it; always known
untimed (fifth state)	bez vremena (peto stanje)	value exact, age unknown; drawn with an age band fading left
absence	odsustvo	a recorded fact that nothing came — row, receipt or slot; never zero
quorum bar	kvorum traka	counted cells: who of those expected actually spoke
permission capture	hvatanje dozvole	the source's robots.txt, headers and licence page stored with hashes before any collector
opted out	odbio	a publisher that said no in a file; never proposed again
dark instrument	tamni instrument	a device that measures and publishes nothing machine-readable
organ	organ	a small local model with a register entry; derives, never measures
monologue	monolog	the running bilingual sentences built from receipts, silences and labelled organ output
pulse	puls	a ring on the map only when an instrument was actually read
TRAKA / KARTON / PRESEK	traka / karton / presek	the three compositions: time, instrument, section
editor of record	odgovorni urednik	the named person who accepts or strikes every model-written public sentence